

BIKRANT NATH

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PROFESSIONAL SUMMARY

M.Tech Software Engineer who cut privacy-attack leakage to near-zero (MIA ~ 0.50) while outperforming two published baselines on AUC under matched evaluation conditions -- built across the full SDLC (design, implementation, testing, maintenance) in Agile team settings. Full-stack delivery background: 5+ production and research systems shipped end-to-end using design patterns, code review, and Git workflows.

TECHNICAL SKILLS

- **Engineering Practices:** SDLC, Agile/Scrum, Design Patterns (Singleton, Observer, Strategy), Unit Testing, Debugging, Code Review, Git
- **Languages & Frameworks:** Python, JavaScript, Java, C, C#, React, Next.js, Node.js/Express, PyTorch, Opacus
- **AI / ML:** Deep Learning, GANs, Differential Privacy, LLMs, RAG, Prompt Engineering, Hyperparameter Tuning
- **Databases & Other:** MongoDB, SQL, WebSockets, REST APIs, Postman, Nmap/Burp Suite (security testing)

EXPERIENCE

Software Development Intern | Fineworks Media & Solution Technologies

Jan 2022 - Jul 2022 · Assam

- Cut manual booking errors, as measured by fewer double-bookings in QA testing, by building a full-stack hotel booking platform (React, Node/Express, MongoDB) with automated availability checks in an Agile sprint cycle.
- Kept booking data accurate under concurrent load, as measured by consistent live-status accuracy across simultaneous requests, by designing and testing RESTful APIs for booking, inventory, and reservations.
- Shipped a secure payment flow with zero failed-transaction incidents in testing, as measured by end-to-end confirmation success, by integrating a payment gateway and debugging edge cases via code review.

PROJECTS

DP-CTGAN with Rare Minority Sampling -- Privacy-Preserving Synthetic Data Generation

2025-2026

- Achieved near-zero membership-inference leakage, as measured by MIA scores of ~ 0.50 across four real-world datasets, by designing and maintaining a Differentially Private CTGAN pipeline (PyTorch, Opacus) with a formal (eps, delta)-DP guarantee via a Renyi accountant.
- Lifted rare-class synthetic fidelity to an AUPRC of 0.7933 vs. 0.0317 for the PATE-GAN baseline under a fixed training budget at eps=3 on Credit Card Fraud, by integrating StableNVP normalizing-flow scoring and gradient refinement (Lee et al., arXiv:2412.19543).
- Outperformed two published DP baselines (PATE-GAN, DP-CGAN) on AUC and distributional fidelity (KS) under matched evaluation conditions, by benchmarking across four datasets under a leak-proof 3-way train/val/test split.
- Optimized model configs without evaluation leakage, as measured by TSTR AUC/AUPRC on a sealed holdout, by running a unit-tested, three-stage Optuna search and dual MIA auditing (KNN-MIA, LR-MIA).

Tech: PyTorch, Opacus, Optuna, Scikit-learn, CTGAN, Normalizing Flows, Python

Additional Projects

- **VR Cognitive Assessment (Unity, C#, Meta Quest, React Native) -- 2024:** raised quiz comprehension scores vs. a text-based control group by building a VR computer-literacy learning system, validated via a controlled experiment and structured usability testing.
- **AI Chat Agent (Node.js, WebSockets, LLM API) -- 2025:** delivered real-time, persona-adaptive conversations by building an LLM agent with persistent memory that learns a target user's tone and cadence.
- **Animal Rescue Platform (React, Node.js, MongoDB, Cloudinary) -- 2024:** cut NGO case-response time by shipping real-time geolocation-based matching, alert dispatch, and a multi-role admin dashboard.
- **Athaana eCommerce (React, Node.js, MongoDB) -- 2022:** shipped a production grocery storefront for a Nalbari-based startup handling live customer orders end-to-end, from auth through checkout.
- **Vastramaniaa -- Shopify Store (Liquid, JavaScript) -- 2024:** rebuilt product and collection templates for a clothing brand's storefront.

EDUCATION

M.Tech -- Computer Science & Engineering, Tezpur University

2024 - 2026

CGPA: 7.28 / 10

MCA -- Master of Computer Applications, Tripura University

2022 - 2024

CGPA: 7.71 / 10